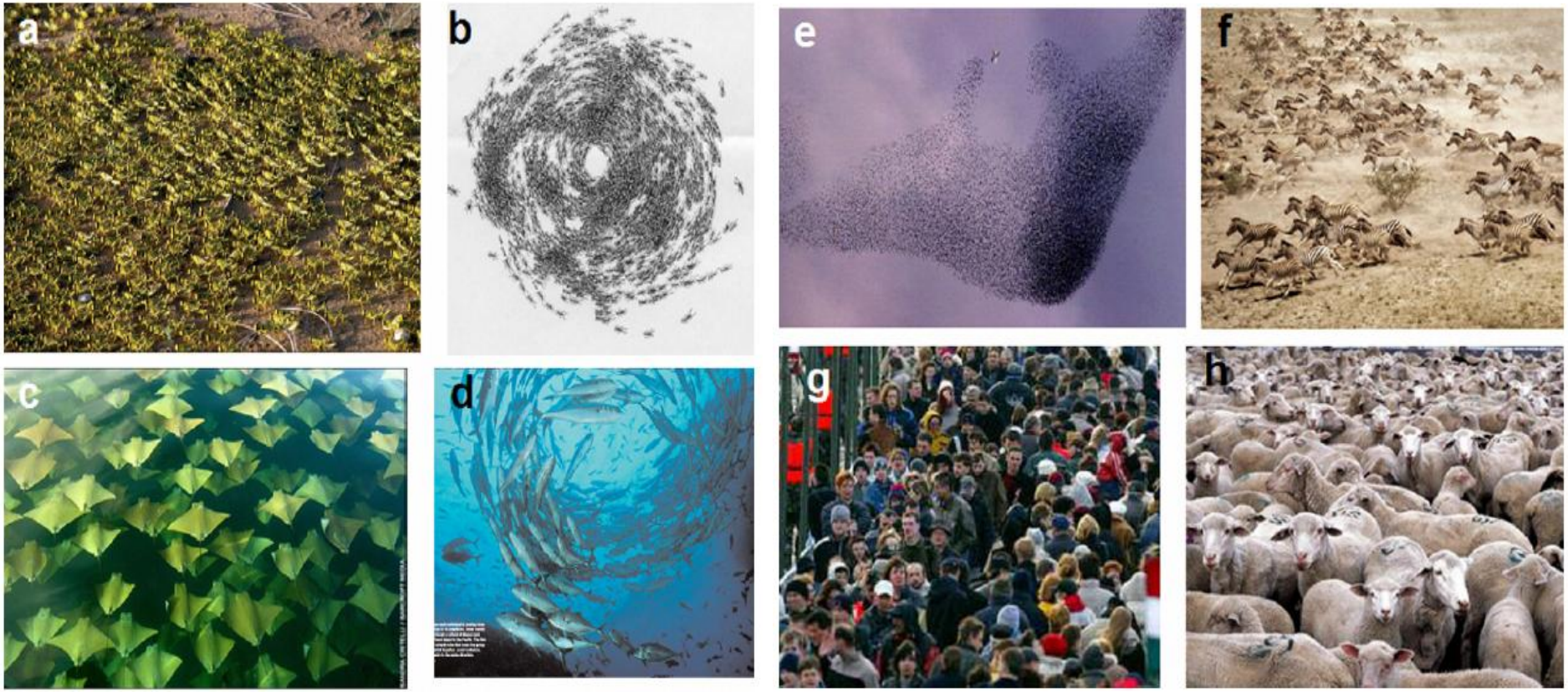
Emergence of Consensus in Self-Organized Collective Motion

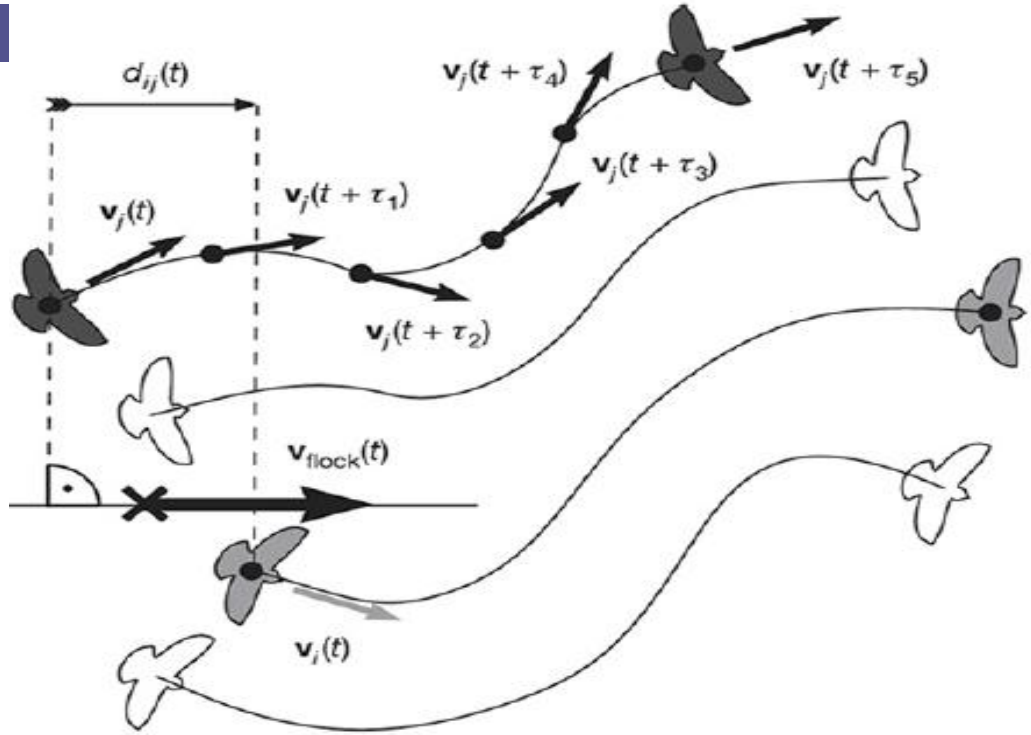
Abstract

Self-organized systems arise very naturally in areas of social, biological, and physical sciences based on alignment. The behavior of such systems is driven solely by interactions among individuals/agents, with the tendency to adjust to each another. This leads to the formation of clusters (e.g., flocks of birds, colonies of ants, and rendezvous by social media). Natural questions that arise in this context are to ask when and how clusters emerge, and what types of rules of engagement influence the formation of such clusters. Of particular interest are cases in which the self-organized dynamics tends to concentrate in one cluster, reflecting a consensus of opinions. The celebrated Cucker-Smale model provides a framework to examine the emergent properties of flocks in order to explain self-organized behaviors in various complex systems. Solutions to Cucker-Smale models describe time evolutions of positions and velocities of agents involved. We have introduced a more realistic approach based on the Cucker-Smale model for self-organized systems with multiple agents. The novelty of our model consists in incorporating processing delays that depend on the distances between agents. Numerical simulations validate our model.

Collective Behavior



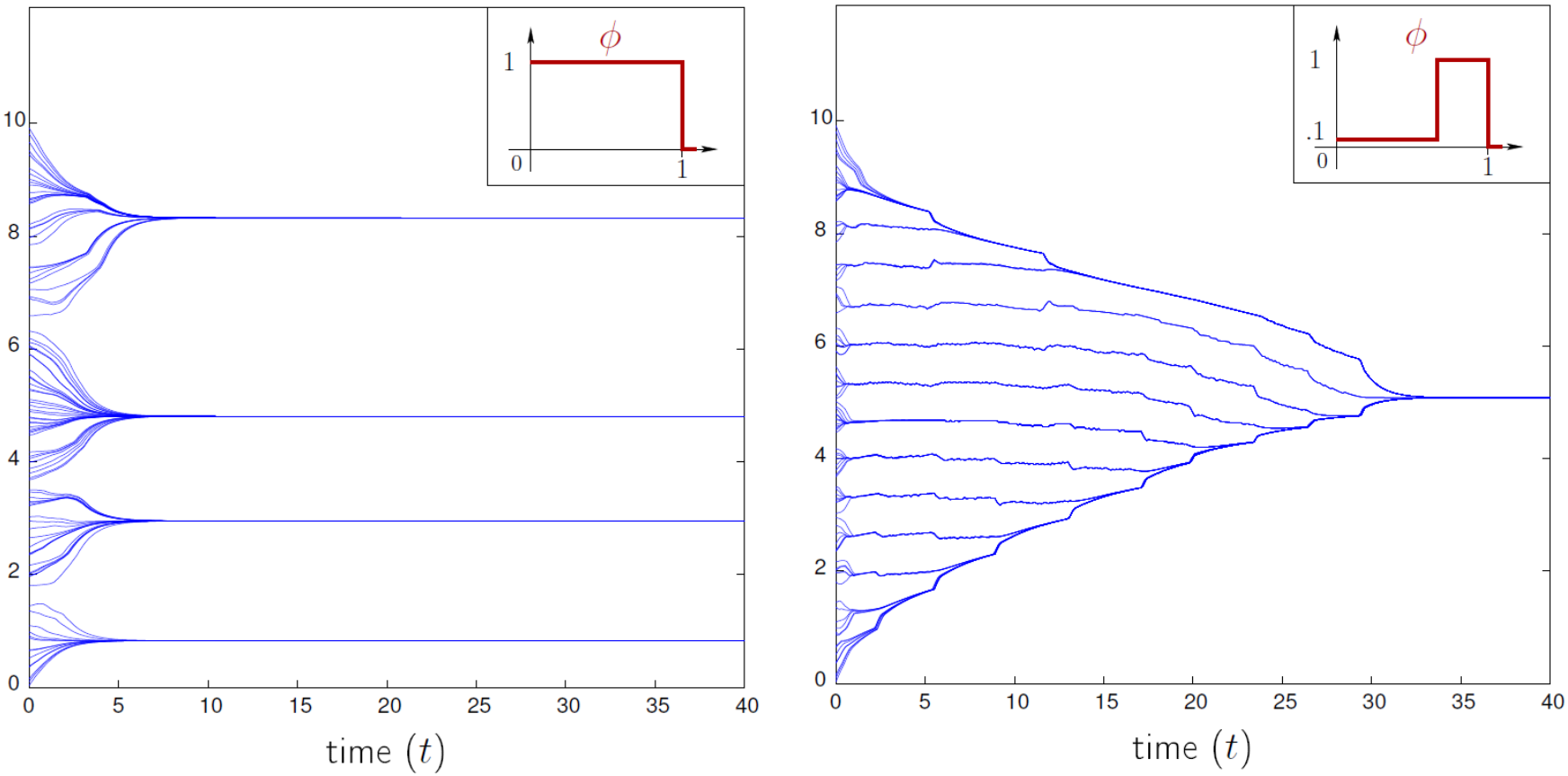
(a) Locusts. (b) Ants. (c) Golden rays. (d) Fish. (e) Starlings. (f) Zebra. (g) People. (h) Sheep.



The classical Cucker-Smale model describes a setting where convergence to consensus in a population of autonomous agents can be formally addressed. Mathematically, it considers the motion of a self-organized group with N agents, with each agent i being characterized by its position $\mathbf{x}_i \in \mathbb{R}^d$ and velocity $\mathbf{v}_i \in \mathbb{R}^d$, where $d \geq 1$ is an integer:

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i, \quad \frac{d\mathbf{v}_i}{dt} = \alpha \sum_{j \neq i} \phi(\|\mathbf{x}_j - \mathbf{x}_i\|) \cdot (\mathbf{v}_j - \mathbf{v}_i).$$

Here, α measures the interaction strength and ϕ is the *influence function* which acts on the difference of positions $d_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$.



Evolution in time for two different influence functions. By diminishing the influence of close neighbors, we enhance the emergence of consensus. Simulations are started with the same initial condition (100 agents uniformly distributed on [0, 10]).

Definition

A flocking solution of the Cucker-Smale model (that describes the evolution of the position and velocity of agents/individuals involved) is a solution for which the agents asymptotically reach a uniform velocity while remaining in finite distance from each other.

Theorem (Cucker-Smale, 2007)

Let $\phi = \frac{H}{(1+\|x_i - x_j\|^2)^\beta}$ *for some fixed* $H > 0$ *and* $\beta > 0$. *Then, there exists a unique solution* $(\mathbf{x}(t), \mathbf{v}(t))$, *defined for all* $t \in \mathbb{R}$, *with initial conditions* $\mathbf{x}(0) = \mathbf{x}_0$ *and* $\mathbf{v}(0) = \mathbf{v}_0$. *If* $\beta < 1/2$ *then, when* $t \to \infty$ *the velocities* $\mathbf{v}_i(t)$ *tend to a common limit* $\hat{\mathbf{v}}$ *and the vectors* $\mathbf{x}_i - \mathbf{x}_j$ *tend to a limit vector* $\hat{\mathbf{x}}_{ij}$, *for all* $i, j \leq N$. *The same happens if* $\beta \geq 1/2$ *provided the initial values* $\mathbf{x}_0$ *and* $\mathbf{v}_0$ *satisfy a given, explicit, relation.*

In 2014, Y. Liu and J. Wu considered the following Cucker-Smale model with constant time delay:

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i, \quad \frac{d\mathbf{v}_i}{dt} = \alpha \sum_{j \neq i} a_{ij}(\mathbf{x}(t-\tau)) \cdot (\mathbf{v}_j(t-\tau) - \mathbf{v}_i),$$

where τ denotes the communication time between agents i and j. To specify a solution for the above self-organized system, one needs to specify the initial conditions

$$\mathbf{x}_i(t) = \mathbf{f}_i(t), \ \mathbf{v}_i(t) = \mathbf{g}_i(t), \quad \text{for } t \in [-\tau, 0],$$

where $\mathbf{f}_i$ and $\mathbf{g}_i$ are given continuous vector-valued functions.

Theorem (Liu-Wu, 2014)

If the influence coefficients a_{ij} satisfy certain technical conditions, then the solution $\{(\mathbf{x}_i(t), \mathbf{v}_i(t)\}_{i=1}^N$ is a flocking solution.

Proposed Model

We are interested in a more general setting by incorporating *delay arguments* in the pairwise influence due to the finite speed in processing the influence. It seems to be very natural to introduce state-dependent time lags for most self-organized systems. Our preliminary results show that these time lags do not change the unconditional flocking property qualitatively, but alter the flocking velocity in a nonlinear way. In general, the influence of agents on each other is realized in various fashions including smell, sound, and vision. For example, the influence among honey bees is transferred mainly by a certain chemical material, while the influence among geese is mainly made through vision. As such, the influence of an agent on another is naturally transferred with a finite speed, and dependent on their relative positions. We have focused in this project on the case of delayed processing of the information about the location and velocity of neighboring agents, resulting in a novel Cucker-Smale model with state-dependent delay.

Let us consider the motion of a self-organized group of N agents, with each agent i characterized by its position $\mathbf{x}_i \in \mathbb{R}^d$ and velocity $\mathbf{v}_i \in \mathbb{R}^d$, where $d \geq 1$ is an integer. Our model is an extension of the Cucker-Smale model in the sense that it includes state-dependent delay:

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i, \quad \frac{d\mathbf{v}_i}{dt} = \alpha \sum_{j \neq i}^{N} a_{ij}(d_{ij}(t - \tau d_{ij}(t)))[\mathbf{v}_j(t - \tau d_{ij}(t)) - \mathbf{v}_i(t)],$$

where α means the interaction strength, the functions a_{ij} quantify the pairwise influence of agent j on the alignment of agent i as a function of the (euclidean) distance $d_{ij}(t) = \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\|$, and $\tau > 0$ is a positive constant involved in the delay argument $t - \tau d_{ij}(t)$.

We also need to specify the initial conditions

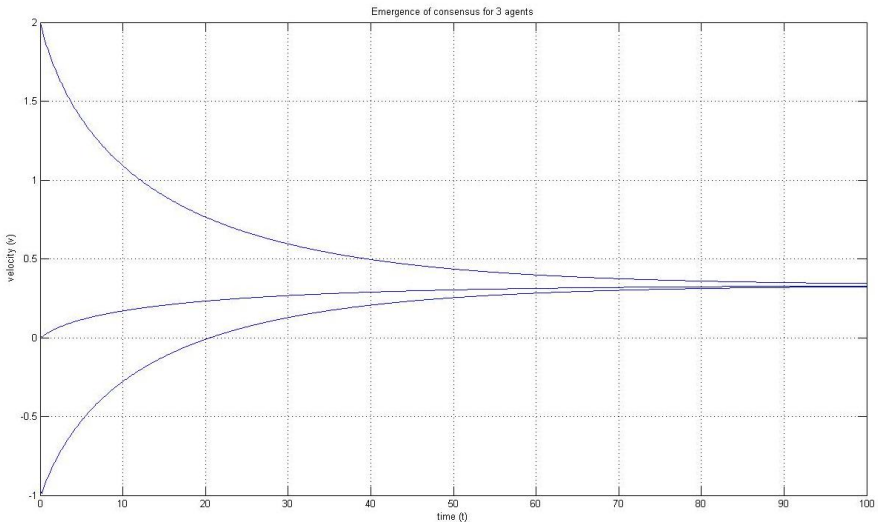
$$\mathbf{x}_i(t) = \mathbf{f}_i(t), \quad \mathbf{v}_i(t) = \mathbf{g}_i(t), \quad \text{for } -\gamma \leq t \leq 0,$$

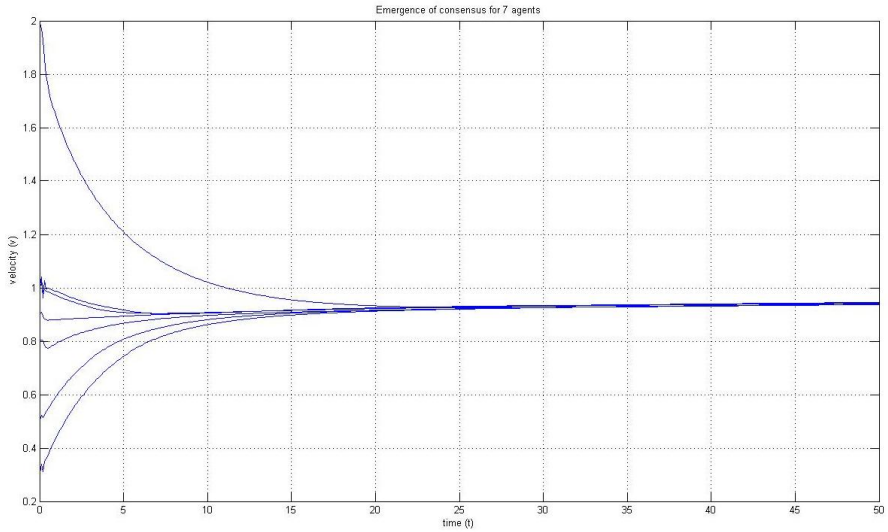
where $\gamma = \min_{i,j}\{\min_{t \geq 0}(t - \tau d_{ij}(t))\}$, and cannot be determined a priori.

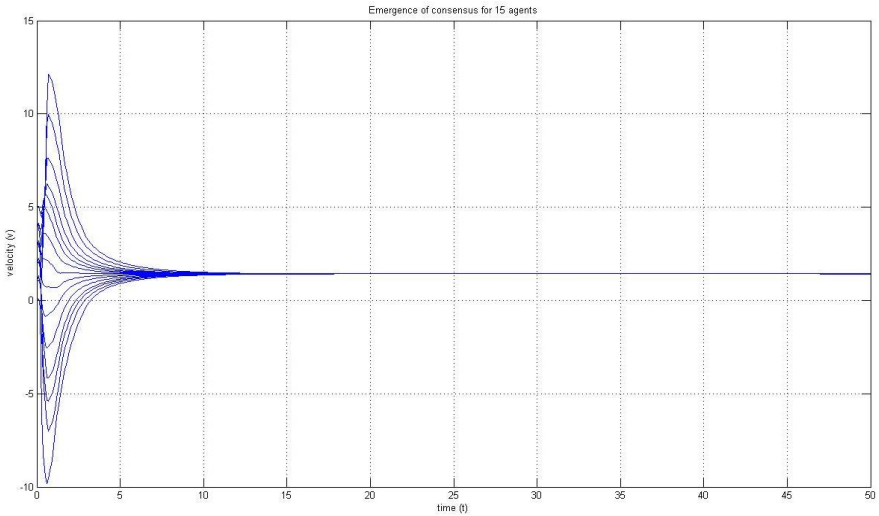
MATLAB Code

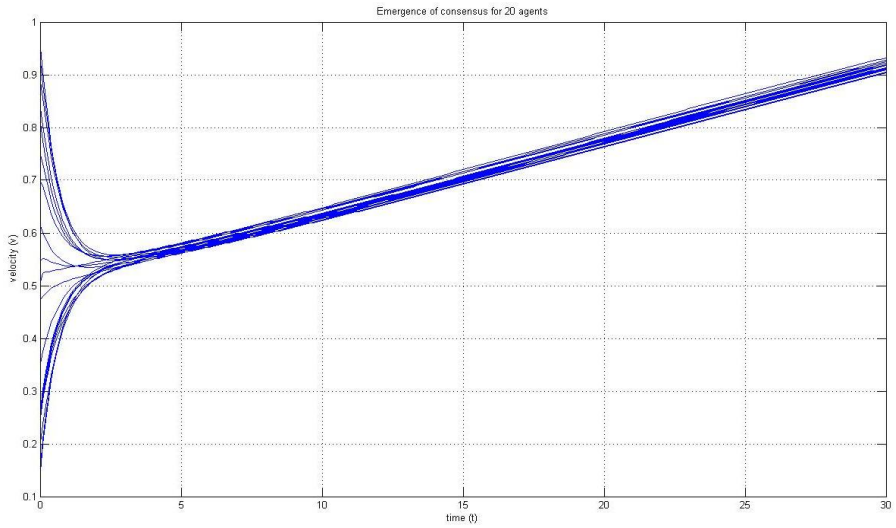
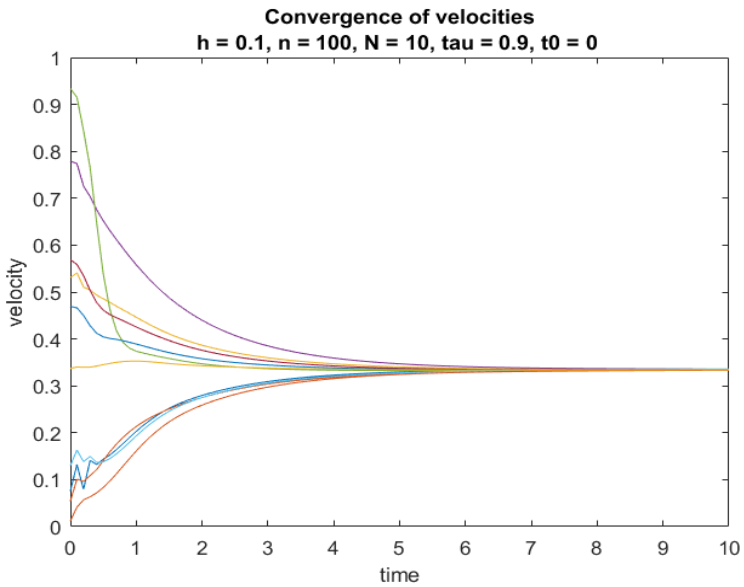
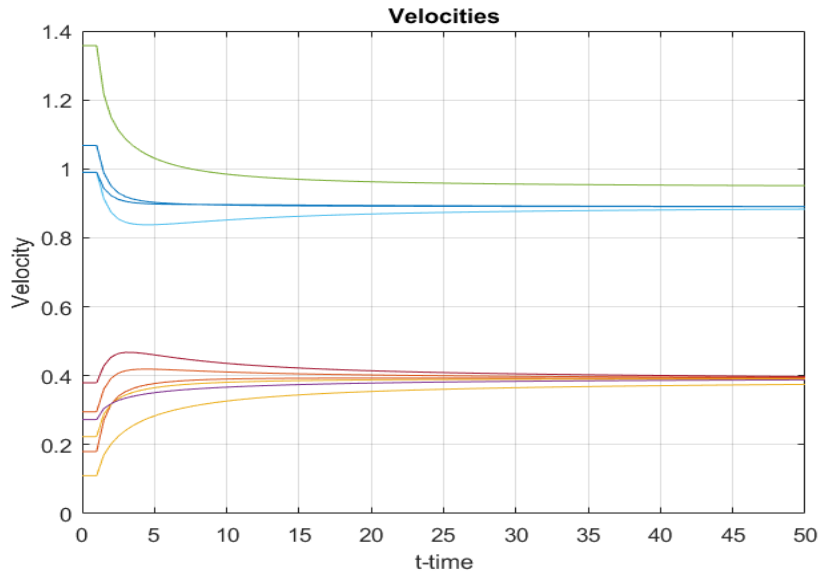
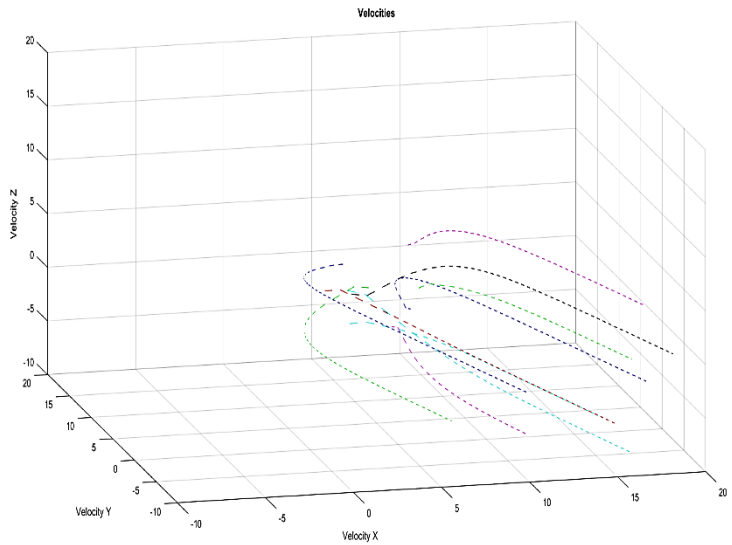
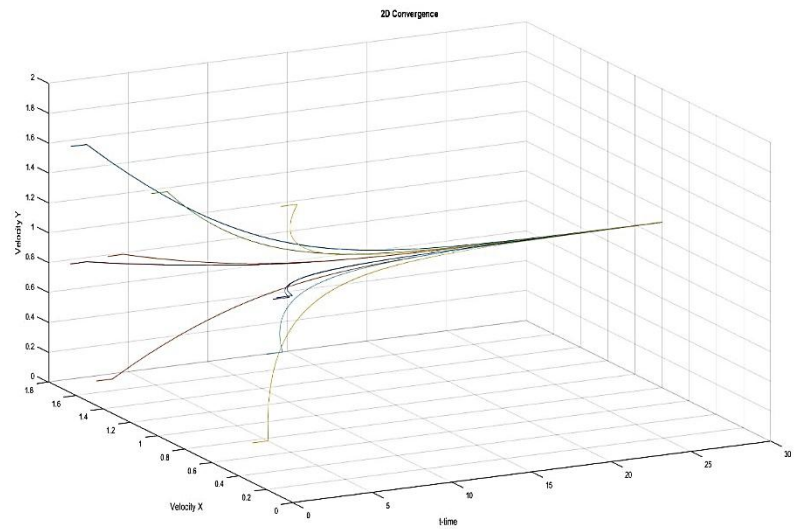
```
h=input('Step size= ')
n=input('Number of steps= ')
N=input('Number of particles= ')
tau=input('Delay coefficient= ')
t0=input('Initial time= ')
t=t0:h:t0+n*h;
x=zeros(N,n);
v=zeros(N,n);
for j=1:N
    disp(['Initial position of particle ',num2str(j),' is'])
    x(j,1)=input('Position= ');
    disp(['Initial velocity of particle',num2str(j),'is'])
    v(j,1)=input('Velocity= ');
end
for i=1:n
    for j=1:N
       x(j,i+1)=x(j,i)+h*v(j,i);
       v(j,i+1)=v(j,i);
       for k=1:N
       v(j,i+1)=v(j,i+1)+h*0.1/(1+(abs(spline(t(1:i+1),x(j,1:i+1)-...
           x(k,1:i+1),t(i)-tau*abs(x(j,i)-x(k,i))))^(1/2)))*...
           (spline(t(1:i+1),v(k,1:i+1),t(i)-tau*abs(x(j,i)-x(k,i)))-v(j,i));
       end
    end
end
for j=1:N
    plot(t,v(j,:))
    hold on
end
hold off
```

Numerical Experiments









Numerical Experiments